

IVAILO, Bulgaria

WMO: 156280

Lat: 42.217N

Lon: 24.333E

Elev: 214

StdP: 98.78

Time Zone: 2.00 (EUE)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.8	-7.7	-13.3	1.2	-9.4	-10.8	1.5	-6.7	13.5	2.4	11.5	4.3	0.5	270	0.436

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	34.6	21.4	32.8	21.1	31.2	20.5	22.9	31.6	22.0	30.4	21.2	29.4	1.7	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.1	15.1	26.8	19.2	14.4	25.7	18.5	13.7	24.8	68.9	31.9	65.4	30.5	62.4	29.5	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	8.3	6.5	-15.0	38.1	4.0	2.3	-17.9	39.7	-20.3	41.1	-22.6	42.4	-25.5	44.1		
(5)			-15.3	24.9	4.0	1.5	-18.2	26.0	-20.6	26.9	-22.9	27.7	-25.8	28.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.6	1.1	3.2	7.4	12.0	17.3	21.3	23.7	23.5	18.8	13.2	7.2	1.9
	DBStd	8.82	4.58	4.71	4.14	3.39	3.05	2.97	2.45	2.43	2.98	3.88	4.03	4.44
	HDD10.0	959	277	192	101	18	0	0	0	0	0	16	103	252
	HDD18.3	2589	534	423	338	192	56	8	1	1	29	164	335	509
	CDD10.0	1909	1	3	22	77	227	339	426	417	264	113	18	2
	CDD18.3	499	0	0	0	1	24	98	168	160	43	4	1	0
	CDH23.3	5371	0	0	3	26	321	1048	1813	1649	452	58	2	0
CDH26.7	2190	0	0	0	4	72	406	824	738	138	8	0	0	
Wind	WSAvg	1.6	1.6	2.0	2.2	1.8	1.6	1.8	1.8	1.4	1.4	1.2	1.3	1.4
Precipitation	PrecAvg	587	46	40	45	44	70	58	55	41	42	45	37	63
	PrecMax	821	92	100	122	86	166	126	156	134	94	113	113	131
	PrecMin	315	7	3	6	14	24	21	0	0	11	3	1	16
	PrecStd	124	21	25	27	22	38	29	39	33	24	27	27	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.7	18.6	22.7	26.5	31.0	34.9	37.5	36.9	32.7	27.8	21.9	16.0
		MCWB	9.4	10.9	13.5	16.8	19.7	21.7	21.5	21.8	20.0	18.2	14.5	11.4
	2%	DB	11.6	15.6	20.1	23.2	28.6	32.7	34.8	34.5	30.2	24.8	18.6	11.9
		MCWB	7.2	9.6	12.0	14.7	18.5	21.3	21.6	21.4	19.3	16.9	13.0	8.7
	5%	DB	9.2	13.0	17.6	21.1	26.8	30.8	32.9	32.5	28.1	22.4	15.6	9.6
		MCWB	6.2	8.5	10.7	13.6	17.7	20.5	21.3	21.0	18.6	15.5	11.3	7.7
	10%	DB	7.1	10.1	15.2	19.2	24.9	29.0	31.1	30.8	25.9	20.0	13.3	7.9
		MCWB	4.8	6.5	9.5	12.7	17.0	19.8	20.5	20.4	17.5	14.4	10.2	6.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.6	11.4	14.3	17.4	20.8	23.6	23.8	24.1	21.1	18.8	15.4	11.9
		MCDB	14.2	17.4	21.2	24.9	28.4	31.4	33.8	32.7	29.7	26.2	21.0	14.5
	2%	WB	7.8	9.8	12.4	15.5	19.6	22.4	22.8	22.9	20.0	17.4	13.5	9.4
		MCDB	10.5	15.1	18.9	22.0	26.9	30.4	31.8	31.7	28.2	23.6	17.2	11.0
	5%	WB	6.4	8.4	11.1	14.3	18.5	21.4	21.9	21.8	19.1	16.2	12.1	8.0
		MCDB	8.7	12.7	16.8	19.7	25.1	29.2	30.4	30.3	26.7	20.8	14.8	9.7
	10%	WB	5.1	6.8	9.7	13.2	17.5	20.4	21.2	20.9	18.2	15.1	10.6	6.5
		MCDB	7.0	9.6	14.7	18.1	23.8	27.7	29.5	29.0	24.4	18.7	12.9	7.8
Mean Daily Temperature Range	5% DB	MDBR	7.3	8.9	10.8	11.6	12.4	13.0	13.6	13.8	12.4	10.5	8.9	6.8
		MCDBR	11.2	14.5	15.8	15.7	16.3	16.4	17.0	17.0	16.6	14.7	13.0	9.9
		MCWBR	7.5	9.4	8.9	8.4	7.6	6.7	6.4	6.5	7.6	7.8	7.8	7.1
	5% WB	MCDBR	10.3	13.7	14.7	13.6	14.7	15.3	15.3	14.9	14.6	12.9	11.1	8.4
		MCWBR	7.2	9.0	8.5	7.6	7.5	6.8	6.4	6.3	7.0	7.1	7.0	6.3
Clear-Sky Solar Irradiance	taub	0.319	0.349	0.397	0.456	0.449	0.450	0.451	0.459	0.432	0.386	0.348	0.324	
	taud	2.382	2.314	2.229	2.073	2.170	2.204	2.209	2.185	2.230	2.347	2.417	2.401	
	Ebn at Noon	817	844	841	814	831	829	823	803	796	792	775	775	
	Edh at Noon	81	102	125	157	146	142	140	139	124	98	79	74	
All-Sky Solar Radiation	RadAvg	1.68	2.49	3.51	4.52	5.46	6.04	6.33	5.76	4.30	2.93	1.93	1.46	
	RadStd	0.24	0.32	0.33	0.51	0.42	0.51	0.47	0.44	0.33	0.33	0.18	0.21	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	+0.75	N/A	N/A	N/A	N/A
(47)	Regional (32 neighbors)	+0.46	N/A	N/A	N/A	+0.60	+0.61	-109	-156	N/A	N/A

Nomenclature: See separate page